

COMPLETE CHEMISTRY

Upper Secondary / Terminale Coursebook

Atomic Structure • Bonding • Reactions • Equilibrium • Organic Chemistry

30 Complete Lessons • Practice Activities • Revision Tests • Answer Key

A structured 80-page learning book for classroom teaching,
independent study, and exam preparation.

GETTING STARTED

How to Use This Book

A practical roadmap for mastering upper-secondary chemistry step by step.

What You Will Learn

- Core chemistry from matter and atomic structure to equilibrium, electrochemistry, organic chemistry, analysis, and environmental chemistry.
- Accurate calculation methods for moles, stoichiometry, gases, solutions, pH, energy, and quantitative analysis.
- Scientific reasoning through model examples, tables, equations, guided practice, revision tests, and answer keys.

Recommended Study Method

- Read the lesson objectives before studying the concepts.
- Study each model equation and explain what every symbol and unit means.
- Complete guided practice without looking at the answers.
- Check your work and write the chemical principle behind every mistake.
- Review difficult lessons after 24 hours and again after one week.

Chemistry Problem-Solving Checklist

- What is given, what is required, and which units are used?
- Is the chemical equation balanced before using mole ratios?
- Are temperature, pressure, volume, and concentration units consistent?
- Does the sign of energy, charge, pH, or oxidation number make chemical sense?
- Have significant figures and realistic magnitude been checked?

Book Organization

- Lessons 1-8 build foundations: matter, atoms, electrons, periodicity, bonding, geometry, and nomenclature.
- Lessons 9-15 develop quantitative chemistry: moles, stoichiometry, gases, solutions, and solubility.
- Lessons 16-22 focus on energy, rates, equilibrium, acids-bases, redox, and electrochemistry.
- Lessons 23-30 complete the course with organic, polymers, analysis, nuclear, environmental chemistry, and review.

Suggested 6-Week Study Schedule

- Week 1: Lessons 1-5 - foundations and atomic structure.
- Week 2: Lessons 6-10 - bonding, nomenclature, moles, stoichiometry.
- Week 3: Lessons 11-15 - yields, states, gases, solutions, solubility.
- Week 4: Lessons 16-20 - thermochemistry, rates, equilibrium, acids and buffers.
- Week 5: Lessons 21-25 - redox, electrochemistry, and organic chemistry.
- Week 6: Lessons 26-30 + all revision tests and error review.

REFERENCE

Table of Contents

Lessons 1-15

Lesson	Topic	Pages
1	Matter and Measurement	5-6
2	Atomic Structure	7-8
3	Isotopes and Atomic Mass	9-10
4	Electron Configuration	11-12
5	Periodic Table and Trends	13-14
6	Chemical Bonding	15-16
7	Molecular Geometry and Polarity	17-18
8	Chemical Nomenclature and Formulae	19-20
9	Mole Concept and Molar Mass	21-22
10	Stoichiometry	23-24
11	Limiting Reagents and Percent Yield	25-26
12	States of Matter and Intermolecular Forces	27-28
13	Gas Laws	29-30
14	Solutions and Concentration	31-32
15	Solubility and Colligative Properties	33-34

Part I Learning Map

- Lessons 1-5: Matter, atomic structure, isotopes, electrons, and periodic trends.
- Lessons 6-10: Bonding, molecular geometry, nomenclature, mole concept, and stoichiometry.
- Lessons 11-15: Limiting reagents, states, gases, solutions, and solubility.

Part I Milestones

- I can connect atomic structure with periodic properties.
- I can distinguish bonding types and write chemical names/formulae.
- I can solve mole and stoichiometry calculations with balanced equations.
- I can use gas, solution, and solubility concepts accurately.

Part I Study Notes

Use this section to review the concepts you need to review and the principle you want to remember.

REFERENCE

Table of Contents

Lessons 16-30, Revision and Reference

Lesson	Topic	Pages
16	Thermochemistry	35-36
17	Reaction Rates	37-38
18	Chemical Equilibrium	39-40
19	Acids, Bases and pH	41-42
20	Acid-Base Equilibria and Buffers	43-44
21	Oxidation-Reduction Reactions	45-46
22	Electrochemistry	47-48
23	Organic Chemistry Foundations	49-50
24	Hydrocarbons	51-52
25	Functional Groups and Organic Reactions	53-54
26	Polymers and Biomolecules	55-56
27	Analytical Chemistry: Qualitative Methods	57-58
28	Analytical Chemistry: Quantitative Methods	59-60
29	Nuclear Chemistry	61-62
30	Environmental Chemistry and Course Review	63-64
Revision	Comprehensive Review Tests	65-72
Answers	Answer Key	73-78
Reference	Periodic Table, Constants & Ions	79
Reference	Chemistry Quick Guide	80

Part II Learning Map

- Lessons 16-20: Energy, rates, equilibrium, acids-bases, and buffers.
- Lessons 21-25: Redox, electrochemistry, organic foundations, hydrocarbons, and functional groups.
- Lessons 26-30: Polymers, analysis, nuclear chemistry, environmental chemistry, and integration.

Part II Milestones

- I can explain energy changes, kinetics, and equilibrium shifts.
- I can solve core acid-base, redox, and electrochemistry problems.
- I can recognize major organic functional groups and polymer concepts.
- I can interpret qualitative/quantitative evidence and environmental chemistry issues.

Part II Study Notes

Study notes before completing the eight review tests.

LESSON 01 - CORE CHEMISTRY

Matter and Measurement

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Classify matter as pure substances or mixtures.
- Use SI units, scientific notation, and significant figures.
- Distinguish accuracy, precision, and uncertainty in measurements.

Concept	Explanation	Example
Matter	Anything that has mass and occupies space.	air, water, copper
Pure substances	Elements contain one kind of atom; compounds contain fixed ratios of elements.	Cu; H ₂ O
Mixtures	Physical combinations with variable composition; homogeneous or heterogeneous.	salt solution; soil
Measurement	A number must be reported with a unit and appropriate significant figures.	12.50 mL

Key Rules

- Convert units using dimensional analysis.
- In multiplication/division, round by significant figures; in addition/subtraction, round by decimal places.
- A measured value should never imply more precision than the instrument provides.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 1.

Extra Model Examples

- Density = mass / volume
- 2.50 g/cm³ means 2.50 grams per cubic centimetre.
- 1.25 km = 1.25 x 10³ m
- A homogeneous mixture has uniform composition throughout.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 01 - PRACTICE

Matter and Measurement

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
$5\text{ cm} + 2\text{ g} = 7$	Do not add quantities with different dimensions.	Units must be compatible.
0.00450 has two significant figures.	0.00450 has three significant figures.	Leading zeros are not significant; trailing decimal zeros are.

Guided Practice	
1. Classify oxygen gas: element, compound, or mixture.	2. Classify salt water as homogeneous or heterogeneous.
3. Convert 2500 mg to g.	4. Write 0.00072 in scientific notation.
5. How many significant figures are in 12.030?	6. Calculate density for 20.0 g occupying 5.0 mL.
7. Explain the difference between accuracy and precision.	8. Name one SI base unit used in chemistry.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. element	2. homogeneous mixture
3. 2.500 g	4. 7.2×10^{-4}
5. five	6. 4.0 g/mL
7. Accuracy is closeness to the accepted value; precision is repeatability.	8. Example: metre, kilogram, second, kelvin, mole.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 02 - CORE CHEMISTRY

Atomic Structure

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Describe protons, neutrons, and electrons.
- Relate atomic number and mass number to atomic composition.
- Explain the basic structure of the atom.

Concept	Explanation	Example
Proton	Positively charged particle in the nucleus.	charge +1; mass about 1 u
Neutron	Neutral particle in the nucleus.	charge 0; mass about 1 u
Electron	Negatively charged particle outside the nucleus.	charge -1; very small mass
Atomic number	Number of protons; defines the element.	Carbon has $Z = 6$

Key Rules

- For a neutral atom: protons = electrons.
- Mass number A = protons + neutrons.
- For ions, electron number changes but proton number does not.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 2.

Extra Model Examples

- Na: $Z = 11 \rightarrow 11$ protons
- ^{23}Na has 12 neutrons because $23 - 11 = 12$.
- Na^+ has 10 electrons.
- The nucleus contains nearly all atomic mass.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 02 - PRACTICE

Atomic Structure

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
An ion changes its number of protons.	An ion changes its number of electrons.	Changing protons changes the element.
Mass number equals average atomic mass.	Mass number is for one isotope; average atomic mass is weighted.	They represent different quantities.

Guided Practice	
1. How many protons are in oxygen ($Z=8$)?	2. How many neutrons are in ^{16}O ?
3. How many electrons are in Mg^{2+} ($Z=12$)?	4. Define atomic number.
5. Define mass number.	6. Why is the atom electrically neutral when protons equal electrons?
7. Where are neutrons found?	8. What particle determines chemical identity?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. 8	2. 8
3. 10	4. Number of protons.
5. Protons + neutrons.	6. Positive and negative charges cancel.
7. In the nucleus.	8. The proton number / atomic number.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

LESSON 03 - CORE CHEMISTRY

Isotopes and Atomic Mass

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Define isotopes and nuclide notation.
- Calculate weighted average atomic mass.
- Interpret natural abundance data.

Concept	Explanation	Example
Isotopes	Atoms of the same element with different neutron numbers.	^{12}C , ^{13}C , ^{14}C
Nuclide notation	Mass number and atomic number identify a specific isotope.	^A_ZX notation
Natural abundance	Fraction or percentage of each isotope in a natural sample.	75% isotope A
Average atomic mass	Weighted mean of isotope masses.	$\text{sum}(\text{mass} \times \text{fraction})$

Key Rules

- Isotopes have the same chemical identity but different masses.
- Convert percentages to decimal fractions before weighted averaging.
- Average atomic mass can be non-integer because it is a weighted mean.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 3.

Extra Model Examples

- If isotopes are 10 u (20%) and 11 u (80%), average = 10.8 u.
- ^{35}Cl and ^{37}Cl are isotopes of chlorine.
- Isotope abundance fractions sum to 1.
- Radioisotopes have unstable nuclei.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 03 - PRACTICE

Isotopes and Atomic Mass

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Isotopes have different proton numbers.	Isotopes have the same proton number but different neutron numbers.	Proton number defines the element.
Average mass must be a whole number.	Average mass can be decimal.	It is a weighted average.

Guided Practice	
1. What differs between isotopes of one element?	2. Calculate neutrons in ^{37}Cl , $Z=17$.
3. If isotope A is 20 u at 25% and B is 22 u at 75%, find the average.	4. Why do isotopes occupy the same place in the periodic table?
5. Write the abundance fraction for 18%.	6. Do isotopes have identical masses?
7. What is a radioisotope?	8. Why is average atomic mass usually not an integer?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Number of neutrons.	2. 20
3. 21.5 u	4. They have the same atomic number.
5. 0.18	6. No.
7. An isotope with an unstable nucleus.	8. Because it averages several isotopes by abundance.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

LESSON 04 - CORE CHEMISTRY

Electron Configuration

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Describe shells, subshells, and orbitals qualitatively.
- Write electron configurations for common atoms and ions.
- Relate valence electrons to chemical behaviour.

Concept	Explanation	Example
Energy levels	Electrons occupy quantized energy levels around the nucleus.	$n = 1, 2, 3...$
Subshells	Common subshells are s, p, d, and f.	2p, 3s
Orbitals	An orbital holds a maximum of two electrons.	1s ²
Valence electrons	Outer electrons involved most directly in bonding.	Na has one valence electron

Key Rules

- Fill lower-energy orbitals before higher-energy orbitals in the basic Aufbau pattern.
- Maximum capacities: s=2, p=6, d=10, f=14.
- For main-group ions, electrons are gained or lost to reach more stable arrangements.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 4.

Extra Model Examples

- O: 1s² 2s² 2p⁴
- Na: 1s² 2s² 2p⁶ 3s¹
- Na⁺: 1s² 2s² 2p⁶
- Valence electrons help predict group chemistry.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 04 - PRACTICE

Electron Configuration

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
2p can hold eight electrons.	2p can hold six electrons.	A p subshell contains three orbitals.
Na ⁺ has one more electron than Na.	Na ⁺ has one fewer electron than Na.	Positive ions form by electron loss.

Guided Practice	
1. Write the configuration of He.	2. Write the configuration of C.
3. How many electrons fit in an s subshell?	4. How many fit in a p subshell?
5. How many valence electrons does Na have?	6. What happens to electrons when Cl becomes Cl ⁻ ?
7. Why are valence electrons important?	8. Name the four common subshell labels.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. 1s ²	2. 1s ² 2s ² 2p ²
3. 2	4. 6
5. 1	6. It gains one electron.
7. They govern bonding and reactivity.	8. s, p, d, f

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections You made, the correction, and the chemistry rule or principle.

LESSON 05 - CORE CHEMISTRY

Periodic Table and Trends

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Use groups and periods to locate elements.
- Explain basic periodic trends.
- Predict metallic and nonmetallic character.

Concept	Explanation	Example
Groups	Vertical columns; main-group members have related valence patterns.	Group 1 alkali metals
Periods	Horizontal rows corresponding broadly to occupied principal levels.	Period 3
Atomic radius	Generally decreases across a period and increases down a group.	K larger than Na
Ionization energy	Energy needed to remove an electron; generally increases across a period.	higher toward upper right

Key Rules

- Periodic trends are general patterns, not absolute without exceptions.
- Metals dominate the left and centre; nonmetals dominate the upper right.
- Electronegativity generally increases across a period and decreases down a group.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 5.

Extra Model Examples

- Na and K have similar group chemistry.
- Cl is more electronegative than Na.
- K has a larger radius than Li.
- Noble gases are relatively unreactive.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 05 - PRACTICE

Periodic Table and Trends

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Atomic radius always increases left to right.	Atomic radius generally decreases left to right.	Effective nuclear attraction increases across a period.
All elements in a period have identical properties.	Properties change progressively across a period.	Valence structure changes.

Guided Practice	
1. Which direction generally increases atomic radius down a group?	2. Which direction generally increases electronegativity across a period?
3. Name Group 18.	4. Are halogens metals or nonmetals?
5. Which is larger: Na or K?	6. Which is more electronegative: O or S?
7. What is a period?	8. What is a group?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Down.	2. Left to right.
3. Noble gases.	4. Nonmetals.
5. K.	6. O.
7. A horizontal row.	8. A vertical column.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 06 - CORE CHEMISTRY

Chemical Bonding

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Distinguish ionic, covalent, and metallic bonding.
- Relate bonding to electron transfer or sharing.
- Predict broad physical properties from bond type.

Concept	Explanation	Example
Ionic bonding	Electrostatic attraction between oppositely charged ions.	NaCl
Covalent bonding	Atoms share electron pairs.	H ₂ , H ₂ O
Metallic bonding	Positive metal centres are attracted to delocalized electrons.	Cu, Fe
Bond energy	Energy associated with breaking or forming bonds.	stronger bonds need more energy to break

Key Rules

- Ionic compounds usually form between metals and nonmetals.
- Covalent compounds commonly form among nonmetals.
- Bond type influences melting point, conductivity, and solubility.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 6.

Extra Model Examples

- $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$
- $\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$
- H-H represents one covalent bond.
- Copper conducts because electrons are mobile.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 06 - PRACTICE

Chemical Bonding

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Ionic bonding is electron sharing.	Ionic bonding involves attraction after electron transfer.	Sharing describes covalent bonding.
Solid NaCl conducts electricity well.	Molten or aqueous NaCl conducts better than solid NaCl.	Ions must be mobile.

Guided Practice	
1. Identify bond type in NaCl.	2. Identify bond type in H ₂ .
3. Identify bonding in copper metal.	4. Why do metals conduct electricity?
5. What happens to electrons in a covalent bond?	6. What particles attract in an ionic crystal?
7. Name one property typical of ionic solids.	8. Name one property typical of metals.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. ionic	2. covalent
3. metallic	4. Delocalized electrons can move.
5. They are shared.	6. Oppositely charged ions.
7. High melting point / brittle.	8. Conductive / malleable / ductile.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 07 - CORE CHEMISTRY

Molecular Geometry and Polarity

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Use electron-pair repulsion to predict simple shapes.
- Distinguish bond polarity from molecular polarity.
- Relate geometry to intermolecular interactions.

Concept	Explanation	Example
VSEPR idea	Electron regions repel and arrange to minimize repulsion.	CH ₄ tetrahedral
Linear	Two electron regions around a central atom.	CO ₂
Bent	Lone pairs can produce a bent shape.	H ₂ O
Polarity	Overall polarity depends on bond dipoles and molecular symmetry.	H ₂ O polar; CO ₂ nonpolar

Key Rules

- Lone pairs generally repel more strongly than bonding pairs.
- Symmetrical molecules can have polar bonds but no net dipole.
- Geometry matters when predicting boiling points and solubility.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 7.

Extra Model Examples

- CO₂: linear and overall nonpolar.
- H₂O: bent and polar.
- CH₄: tetrahedral and nonpolar.
- NH₃: trigonal pyramidal and polar.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 07 - PRACTICE

Molecular Geometry and Polarity

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Every molecule with polar bonds is polar.	Molecular symmetry can cancel bond dipoles.	Net dipole depends on vector sum.
H ₂ O is linear.	H ₂ O is bent.	Two lone pairs change the geometry.

Guided Practice	
1. State the shape of CO ₂ .	2. State the shape of CH ₄ .
3. Is H ₂ O polar?	4. Why can CO ₂ be nonpolar despite polar bonds?
5. What effect do lone pairs have on shape?	6. State the approximate shape of NH ₃ .
7. Define bond polarity.	8. What determines overall molecular polarity?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. linear	2. tetrahedral
3. yes	4. Its bond dipoles cancel because the molecule is linear and symmetric.
5. They alter electron-pair arrangement and bond angles.	6. trigonal pyramidal
7. Unequal electron sharing in a bond.	8. Bond polarities plus molecular geometry.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 08 - CORE CHEMISTRY

Chemical Nomenclature and Formulae

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Name common ionic and molecular compounds.
- Write formulae from ion charges.
- Recognize acids and simple polyatomic ions.

Concept	Explanation	Example
Ionic names	Name cation first, then anion; variable-charge metals may need Roman numerals.	sodium chloride; iron(III) oxide
Molecular names	Prefixes can show atom counts in binary molecular compounds.	carbon dioxide
Formula writing	Total positive and negative charge must balance.	$\text{Ca}^{2+} + 2\text{Cl}^- \rightarrow \text{CaCl}_2$
Polyatomic ions	Charged groups act as units in formulae.	SO_4^{2-} , NO_3^-

Key Rules

- Do not use prefixes such as di- for simple ionic compounds.
- Use parentheses when more than one polyatomic ion is needed.
- Check charge neutrality after writing an ionic formula.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 8.

Extra Model Examples

- Mg^{2+} and $\text{Cl}^- \rightarrow \text{MgCl}_2$
- Al^{3+} and $\text{O}^{2-} \rightarrow \text{Al}_2\text{O}_3$
- CO_2 = carbon dioxide
- $\text{Ca}(\text{NO}_3)_2$ = calcium nitrate

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 08 - PRACTICE

Chemical Nomenclature and Formulae

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
NaCl is sodium monochloride.	NaCl is sodium chloride.	Simple ionic names do not use molecular prefixes.
Ca ²⁺ and Cl ⁻ gives CaCl.	The formula is CaCl ₂ .	Charges must balance.

Guided Practice	
1. Name Na ₂ O.	2. Name CO ₂ .
3. Write the formula for magnesium chloride.	4. Write the formula for aluminium oxide.
5. Name Ca(NO ₃) ₂ .	6. What is the charge on sulfate?
7. Why are parentheses used in Ca(NO ₃) ₂ ?	8. Write the formula for sodium sulfate.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. sodium oxide	2. carbon dioxide
3. MgCl ₂	4. Al ₂ O ₃
5. calcium nitrate	6. 2-
7. There are two nitrate ions.	8. Na ₂ SO ₄

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 09 - CORE CHEMISTRY

Mole Concept and Molar Mass

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Relate moles, particles, and mass.
- Calculate molar mass from a formula.
- Use Avogadro constant in particle calculations.

Concept	Explanation	Example
Mole	Amount of substance containing Avogadro number of entities.	1 mol = 6.022×10^{23} entities
Molar mass	Mass of one mole, in g/mol.	H ₂ O about 18.0 g/mol
Mass-mole relation	$n = m / M$.	18.0 g H ₂ O is about 1.00 mol
Particles	$N = n \times N_A$.	0.50 mol contains 3.011×10^{23} entities

Key Rules

- Use the formula mass to calculate molar mass.
- Track units: g, mol, g/mol.
- Specify the entity when using Avogadro constant: atoms, molecules, ions, or formula units.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 9.

Extra Model Examples

- $M(\text{H}_2\text{O}) = 2(1.0) + 16.0 = 18.0 \text{ g/mol}$
- $n = 36.0 / 18.0 = 2.00 \text{ mol}$
- $N = 2.00 \times 6.022 \times 10^{23}$
- 1 mol NaCl contains 1 mol Na⁺ and 1 mol Cl⁻ in the formula-unit sense.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create a chemistry example, equation, or calculation and underline the target principle.

LESSON 09 - PRACTICE

Mole Concept and Molar Mass

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Molar mass is measured in moles.	Molar mass is measured in g/mol.	It is mass per amount.
1 mol always means 1 gram.	1 mol has a substance-specific mass.	Mass depends on molar mass.

Guided Practice	
1. What is Avogadro constant?	2. Find molar mass of CO ₂ using C=12.0, O=16.0.
3. Find moles in 44.0 g CO ₂ .	4. Find molecules in 2.0 mol H ₂ O.
5. Find mass of 0.50 mol NaCl if M=58.5 g/mol.	6. State the equation linking moles, mass, and molar mass.
7. What entity is counted for molecular oxygen?	8. What entity is counted for NaCl solid?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. $6.022 \times 10^{23} \text{ mol}^{-1}$	2. 44.0 g/mol
3. 1.00 mol	4. 1.2044×10^{24} molecules
5. 29.25 g	6. $n = m/M$
7. O ₂ molecules	8. formula units

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record corrections you made, the correction, and the chemistry rule or principle.

LESSON 10 - CORE CHEMISTRY

Stoichiometry

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Balance chemical equations.
- Use mole ratios from balanced equations.
- Convert between masses and amounts in reactions.

Concept	Explanation	Example
Balanced equation	Same number of each type of atom on both sides.	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
Mole ratio	Coefficients provide reacting amount ratios.	2 mol H_2 : 1 mol O_2
Stoichiometric calculation	Convert known quantity to moles, apply ratio, then convert to desired unit.	mass $\rightarrow$ mol $\rightarrow$ mol $\rightarrow$ mass
Conservation	Atoms are rearranged, not created or destroyed in ordinary reactions.	mass conserved in closed system

Key Rules

- Never change subscripts to balance an equation.
- Use coefficients from the fully balanced equation.
- Keep units visible through each conversion step.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 10.

Extra Model Examples

- $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- 1 mol CH_4 needs 2 mol O_2 .
- 2 mol H_2 forms 2 mol H_2O .
- Stoichiometric coefficients are exact ratios.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 10 - PRACTICE

Stoichiometry

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Balance by changing H ₂ O to H ₂ O ₂ .	Balance with coefficients only.	Changing subscripts changes the substance.
Use an unbalanced equation for mole ratios.	Balance first.	Ratios come from balanced coefficients.

Guided Practice	
1. Balance: H ₂ + O ₂ → H ₂ O.	2. What is the H ₂ :O ₂ mole ratio in the balanced equation?
3. How many mol H ₂ O form from 3 mol H ₂ if O ₂ is excess?	4. In CH ₄ + 2O ₂ → CO ₂ + 2H ₂ O, how many mol O ₂ are needed for 2 mol CH ₄ ?
5. Why must equations be balanced?	6. Can coefficients be fractional during balancing?
7. What is conserved in chemical reactions?	8. Give the standard three-step stoichiometry path.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. 2H ₂ + O ₂ → 2H ₂ O	2. 2:1
3. 3 mol	4. 4 mol
5. To satisfy conservation of atoms.	6. Yes, but final integer coefficients are usually preferred.
7. Atoms / mass in a closed system.	8. Known quantity → moles → mole ratio → desired quantity.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

LESSON 11 - CORE CHEMISTRY

Limiting Reagents and Percent Yield

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Identify limiting and excess reactants.
- Calculate theoretical yield.
- Calculate percent yield.

Concept	Explanation	Example
Limiting reagent	Reactant consumed first; limits product amount.	smallest product prediction
Excess reagent	Reactant remaining after the limiting reagent is used.	more than stoichiometrically required
Theoretical yield	Maximum product predicted from stoichiometry.	calculated yield
Percent yield	Actual yield / theoretical yield x 100%.	80 g actual / 100 g theoretical = 80%

Key Rules

- Convert each reactant to the same product amount to identify the limiter.
- Theoretical yield is based on the limiting reagent only.
- Percent yield above 100% usually signals measurement, purity, or drying issues.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 11.

Extra Model Examples

- Limiting reagent -> theoretical yield.
- % yield = actual/theoretical x 100.
- Excess amount can be found after subtracting reacted quantity.
- Reaction losses can lower actual yield.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 11 - PRACTICE

Limiting Reagents and Percent Yield

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
The reactant with the smaller mass is always limiting.	Limiting depends on moles and stoichiometric ratio.	Different substances have different molar masses and coefficients.
Percent yield = theoretical/actual x 100.	Percent yield = actual/theoretical x 100.	Actual is compared with the maximum possible.

Guided Practice	
1. Define limiting reagent.	2. Define excess reagent.
3. What determines theoretical yield?	4. Write the percent yield formula.
5. If theoretical yield is 50 g and actual is 40 g, find percent yield.	6. Can the larger-mass reactant be limiting?
7. Name one reason actual yield may be lower than theoretical.	8. What does a yield above 100% often suggest?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. The reactant used up first.	2. Reactant left after the limiting reagent is consumed.
3. The limiting reagent.	4. actual/theoretical x 100%
5. 80%	6. Yes.
7. Incomplete reaction, transfer loss, side reactions, etc.	8. Impurities, retained solvent/water, or measurement error.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 12 - CORE CHEMISTRY

States of Matter and Intermolecular Forces

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Describe solids, liquids, and gases at particle level.
- Distinguish intermolecular forces from chemical bonds.
- Relate forces to boiling point and viscosity.

Concept	Explanation	Example
Solid	Particles are closely packed and mainly vibrate about fixed positions.	crystalline salt
Liquid	Particles remain close but can move past one another.	water
Gas	Particles are far apart and move freely.	air
Intermolecular forces	Attractions between molecules, including dispersion, dipole-dipole, and hydrogen bonding.	H ₂ O hydrogen bonding

Key Rules

- Stronger intermolecular forces usually raise boiling point.
- Hydrogen bonding is especially important when H is bonded to N, O, or F.
- Intermolecular forces are weaker than the covalent bonds within molecules.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 12.

Extra Model Examples

- H₂O has stronger intermolecular forces than CH₄.
- Evaporation occurs at the surface of a liquid.
- Boiling occurs when vapour pressure matches external pressure.
- Dispersion forces act in all molecules.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 12 - PRACTICE

States of Matter and Intermolecular Forces

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Boiling breaks covalent bonds in water molecules.	Boiling mainly overcomes intermolecular attractions.	Molecules remain H ₂ O.
Gases have no particle motion.	Gas particles move rapidly and randomly.	Kinetic motion is central to gas behaviour.

Guided Practice	
1. Which state has a fixed shape?	2. Which state fills its container?
3. Name the strongest common intermolecular force in liquid water.	4. Do dispersion forces occur in nonpolar molecules?
5. Why does stronger attraction often increase boiling point?	6. What happens at boiling?
7. Are intermolecular forces the same as covalent bonds?	8. Which state is most compressible?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. solid	2. gas
3. hydrogen bonding	4. yes
5. More energy is needed to separate molecules.	6. Vapour pressure equals external pressure throughout the liquid.
7. No.	8. gas

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

LESSON 13 - CORE CHEMISTRY

Gas Laws

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Relate pressure, volume, temperature, and amount.
- Use Boyle, Charles, and ideal gas relationships.
- Convert temperatures to kelvin for gas-law calculations.

Concept	Explanation	Example
Boyle law	At constant temperature and amount, pressure is inversely related to volume.	$P_1V_1 = P_2V_2$
Charles law	At constant pressure, volume is proportional to kelvin temperature.	$V_1/T_1 = V_2/T_2$
Combined law	Relates P, V, and T when amount is fixed.	$P_1V_1/T_1 = P_2V_2/T_2$
Ideal gas law	Relates pressure, volume, amount, and temperature.	$PV = nRT$

Key Rules

- Use kelvin, not Celsius, in proportional gas-law equations.
- Keep pressure and volume units consistent with the chosen R.
- Real gases deviate most at high pressure and low temperature.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 13.

Extra Model Examples

- $T(K) = T(C) + 273.15$
- If volume halves at constant T, pressure doubles.
- $PV = nRT$
- 1 atm approximately 101.3 kPa.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 13 - PRACTICE

Gas Laws

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Use 25 directly as T in $PV=nRT$.	Use about 298 K for 25 C.	Absolute temperature is required.
Doubling pressure doubles volume at constant T.	Doubling pressure halves volume at constant T.	Boyle law is inverse.

Guided Practice	
1. Convert 27 C to K.	2. State Boyle law equation.
3. State Charles law relationship.	4. What happens to volume if kelvin temperature doubles at constant pressure?
5. What does R represent?	6. When do real gases deviate most from ideality?
7. If pressure doubles at constant T, what happens to volume?	8. Write the ideal gas law.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. 300.15 K (about 300 K)	2. $P_1V_1=P_2V_2$
3. $V/T = \text{constant}$	4. It doubles.
5. The gas constant.	6. High pressure and low temperature.
7. It halves.	8. $PV=nRT$

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections You made, the correction, and the chemistry rule or principle.

LESSON 14 - CORE CHEMISTRY

Solutions and Concentration

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Distinguish solute and solvent.
- Calculate molarity and dilution.
- Prepare and describe solutions quantitatively.

Concept	Explanation	Example
Solution	Homogeneous mixture of solute in solvent.	NaCl(aq)
Molarity	Moles of solute per litre of solution.	$M = n/V$
Dilution	Adding solvent lowers concentration without changing solute moles.	$M_1V_1 = M_2V_2$
Electrolyte	Produces ions in solution and conducts electricity.	NaCl, HCl

Key Rules

- Use solution volume, not solvent volume, for molarity.
- Convert mL to L before using $M = n/V$.
- Dilution keeps moles of solute constant when no reaction occurs.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 14.

Extra Model Examples

- 0.50 mol in 2.00 L $\rightarrow$ 0.25 M
- 250 mL = 0.250 L
- $M_1V_1 = M_2V_2$
- Sugar solution is typically a nonelectrolyte.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 14 - PRACTICE

Solutions and Concentration

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Molarity unit is mol.	Molarity is mol/L (M).	It is amount per volume.
Use mL directly in $M=n/V$ without unit care.	Convert to litres or use consistent units.	Units must match the definition.

Guided Practice	
1. Define solute.	2. Define solvent.
3. Calculate M for 1.0 mol in 2.0 L.	4. Convert 500 mL to L.
5. Write the dilution equation.	6. What happens to moles of solute during simple dilution?
7. Is NaCl(aq) an electrolyte?	8. Calculate moles in 0.50 L of 2.0 M solution.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Substance dissolved.	2. Medium doing the dissolving.
3. 0.50 M	4. 0.500 L
5. $M_1V_1=M_2V_2$	6. They remain constant.
7. Yes.	8. 1.0 mol

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

LESSON 15 - CORE CHEMISTRY

Solubility and Colligative Properties

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Describe factors affecting solubility.
- Interpret saturated, unsaturated, and supersaturated solutions.
- Explain colligative properties qualitatively.

Concept	Explanation	Example
Solubility	Maximum amount that dissolves under specified conditions.	g solute per 100 g solvent
Saturation	Saturated solution contains the equilibrium maximum at given conditions.	undissolved solid may remain
Temperature effect	Solubility of many solids rises with temperature; gas solubility often falls.	warm soda loses CO ₂ more readily
Colligative properties	Depend mainly on number of dissolved particles.	boiling-point elevation

Key Rules

- Solubility depends on solute-solvent interactions and conditions.
- Pressure strongly affects gas solubility (Henry-type behaviour).
- Electrolytes can have larger particle effects because they form ions.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 15.

Extra Model Examples

- Unsaturated: more solute can dissolve.
- Supersaturated: more dissolved solute than normal equilibrium amount.
- Adding nonvolatile solute lowers vapour pressure.
- Salt can raise the boiling point of water slightly.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 15 - PRACTICE

Solubility and Colligative Properties

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Stirring permanently increases equilibrium solubility.	Stirring mainly increases the rate of dissolving.	Equilibrium solubility is set by conditions.
Gas solubility always increases with temperature.	Gas solubility often decreases as temperature rises.	Warmer liquids release dissolved gas more easily.

Guided Practice	
1. Define saturated solution.	2. Define unsaturated solution.
3. What effect does higher pressure usually have on gas solubility?	4. What effect does heating often have on gas solubility?
5. Name one colligative property.	6. Do colligative properties depend mainly on particle identity or number?
7. What is supersaturation?	8. Does stirring necessarily change final equilibrium solubility?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Contains the equilibrium maximum dissolved solute.	2. Can dissolve more solute.
3. It increases it.	4. It decreases it.
5. Boiling-point elevation / freezing-point depression / vapour-pressure lowering / osmotic pressure.	6. Number.
7. A metastable solution containing more solute than the usual equilibrium amount.	8. No.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 16 - CORE CHEMISTRY

Thermochemistry

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Distinguish exothermic and endothermic changes.
- Use enthalpy and calorimetry relationships.
- Apply Hess law conceptually.

Concept	Explanation	Example
System and surroundings	The system is the process studied; surroundings are everything else.	reaction mixture vs environment
Exothermic	Releases heat to surroundings; $\Delta H < 0$.	combustion
Endothermic	Absorbs heat from surroundings; $\Delta H > 0$.	some dissolutions
Calorimetry	Heat can be estimated with $q = mc\Delta T$.	warming water

Key Rules

- Sign convention matters: negative ΔH for exothermic reactions.
- Hess law allows reaction enthalpies to be added when equations are added.
- Use consistent mass, specific heat, and temperature units in calorimetry.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 16.

Extra Model Examples

- $q = mc\Delta T$
- $\Delta H_{\text{reaction}} = \text{sum of step enthalpies}$
- Combustion usually exothermic.
- Temperature rise of surroundings often signals heat release by system.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 16 - PRACTICE

Thermochemistry

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Exothermic means Delta H positive.	Exothermic means Delta H negative for the system.	The system loses heat.
$q = mc/T$.	$q = mc\Delta T$.	Heat depends on temperature change.

Guided Practice	
1. What sign does Delta H have for an exothermic process?	2. What sign for endothermic?
3. Write the calorimetry equation.	4. What does c represent in $q = mc\Delta T$?
5. State Hess law in words.	6. If water warms during a reaction, what may this suggest about the reaction?
7. Define system.	8. Define surroundings.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. negative	2. positive
3. $q = mc\Delta T$	4. specific heat capacity
5. Overall enthalpy change equals the sum of enthalpy changes of steps.	6. It is likely exothermic.
7. The part of the universe being studied.	8. Everything outside the system.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 17 - CORE CHEMISTRY

Reaction Rates

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Define reaction rate.
- Explain factors affecting rate using collision theory.
- Interpret simple energy-profile ideas.

Concept	Explanation	Example
Reaction rate	Change in concentration or amount per unit time.	mol/L/s
Collision theory	Particles must collide with sufficient energy and suitable orientation.	effective collisions
Activation energy	Minimum energy barrier for a successful reaction pathway.	E _a
Catalyst	Provides an alternative pathway with lower activation energy.	enzyme, metal catalyst

Key Rules

- Higher temperature usually increases the fraction of effective collisions.
- Greater concentration or pressure can increase collision frequency.
- A catalyst changes rate but is not consumed overall and does not change equilibrium constant.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 17.

Extra Model Examples

- Powdered solid reacts faster than a large chunk because surface area is greater.
- Catalyst lowers activation energy.
- Rate often decreases as reactants are consumed.
- Temperature affects both speed and collision energy distribution.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 17 - PRACTICE

Reaction Rates

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
A catalyst increases the reaction enthalpy.	A catalyst lowers activation energy, not Delta H.	Initial and final energies are unchanged.
More surface area slows a solid reaction.	More surface area usually speeds it.	More particles are exposed for collision.

Guided Practice	
1. Define reaction rate.	2. Name two factors that increase reaction rate.
3. What does a catalyst do to activation energy?	4. Does a catalyst get used up overall?
5. Why does powder react faster than a lump?	6. How does higher temperature affect collision success?
7. What is an effective collision?	8. Does a catalyst change equilibrium composition?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Change in amount or concentration per unit time.	2. Examples: temperature, concentration, pressure for gases, surface area, catalyst.
3. Lowers it.	4. No.
5. Greater surface area.	6. More collisions exceed activation energy.
7. A collision with sufficient energy and suitable orientation.	8. No, it speeds forward and reverse processes.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections You made, the correction, and the chemistry rule or principle.

LESSON 18 - CORE CHEMISTRY

Chemical Equilibrium

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Describe dynamic equilibrium.
- Write simple equilibrium expressions.
- Apply Le Chatelier principle qualitatively.

Concept	Explanation	Example
Dynamic equilibrium	Forward and reverse reactions continue at equal rates.	constant macroscopic concentrations
Equilibrium constant	Relates equilibrium concentrations for a given reaction and temperature.	Kc
Le Chatelier principle	A system shifts to oppose an imposed change.	add reactant -> shift toward products, generally
Temperature	Changing temperature changes the equilibrium constant.	heat treated as reactant/product conceptually

Key Rules

- At equilibrium, concentrations are constant but not necessarily equal.
- Catalysts do not change K or equilibrium position; they only help reach equilibrium faster.
- Pure solids and liquids are usually omitted from simple Kc expressions.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 18.

Extra Model Examples

- $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
- Adding H_2 can shift ammonia equilibrium toward products.
- Removing product can drive more product formation.
- A catalyst reduces the time required to reach equilibrium.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 18 - PRACTICE

Chemical Equilibrium

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
At equilibrium the reaction stops.	Forward and reverse reactions continue at equal rates.	Equilibrium is dynamic.
A catalyst increases K.	A catalyst does not change K.	K depends on temperature for a given reaction.

Guided Practice	
1. Define dynamic equilibrium.	2. Are forward and reverse rates equal at equilibrium?
3. Are reactant and product concentrations necessarily equal?	4. What does a catalyst do to equilibrium position?
5. What may happen if a reactant is added?	6. What may happen if a product is removed?
7. What variable changes K for a given reaction?	8. State Le Chatelier principle briefly.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. A state where forward and reverse reactions continue at equal rates.	2. Yes.
3. No.	4. It does not change it.
5. The system tends to shift toward products.	6. The system tends to form more product.
7. Temperature.	8. An equilibrium shifts to oppose a change.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections You made, the correction, and the chemistry rule or principle.

LESSON 19 - CORE CHEMISTRY

Acids, Bases and pH

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Define acids and bases in common frameworks.
- Calculate simple pH and pOH.
- Recognize strong and weak acids/bases.

Concept	Explanation	Example
Arrhenius acid/base	Acids increase H^+ in water; bases increase OH^- .	HCl ; $NaOH$
Bronsted-Lowry	Acid donates H^+ ; base accepts H^+ .	NH_3 accepts H^+
pH	$pH = -\log[H^+]$.	$[H^+] = 1 \times 10^{-3} \text{ M} \rightarrow pH \ 3$
Neutralization	Acid and base react to form salt and often water.	$HCl + NaOH \rightarrow NaCl + H_2O$

Key Rules

- At 25 C, $pH + pOH = 14$ for dilute aqueous solutions.
- Strong refers to degree of ionization, not concentration.
- Weak acids and bases ionize only partially.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 19.

Extra Model Examples

- pH 2 is more acidic than pH 5.
- $[H^+] = 10^{-pH}$
- Strong acid: nearly complete ionization in dilute water.
- Neutralization stoichiometry depends on H^+ and OH^- equivalents.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 19 - PRACTICE

Acids, Bases and pH

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
A concentrated weak acid is always weaker in effect than a dilute strong acid.	Strength and concentration are different concepts.	Strength is ionization tendency; concentration is amount per volume.
Higher pH means more acidic.	Lower pH means more acidic.	pH decreases as $[H^+]$ increases.

Guided Practice	
1. Define a Bronsted acid.	2. Define a Bronsted base.
3. Calculate pH of 1.0×10^{-3} M H^+ solution.	4. What is $[H^+]$ at pH 4?
5. At 25 C, what is pOH if pH=9?	6. Is pH 2 more acidic than pH 5?
7. What does strong acid mean?	8. Write a simple neutralization equation using HCl and NaOH.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. H^+ donor.	2. H^+ acceptor.
3. 3	4. 1.0×10^{-4} M
5. 5	6. Yes.
7. It ionizes almost completely in water.	8. $HCl + NaOH \rightarrow NaCl + H_2O$

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 20 - CORE CHEMISTRY

Acid-Base Equilibria and Buffers

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Use K_a and K_b concepts qualitatively.
- Explain buffers and common-ion effects.
- Interpret titration curves conceptually.

Concept	Explanation	Example
Weak-acid equilibrium	Weak acids establish reversible ionization equilibria.	$HA \rightleftharpoons H^+ + A^-$
K_a	Equilibrium constant describing acid ionization.	larger $K_a \rightarrow$ stronger weak acid
Buffer	Mixture that resists pH change when small acid/base amounts are added.	weak acid + conjugate base
Titration	Controlled reaction used to determine concentration.	acid-base titration

Key Rules

- Buffers require significant amounts of a weak species and its conjugate partner.
- At the equivalence point, stoichiometric acid-base amounts have reacted.
- Indicators change colour over a characteristic pH range and should suit the titration.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 20.

Extra Model Examples

- CH_3COOH / CH_3COO^- is a common buffer pair.
- Adding small H^+ to a buffer is consumed by the base component.
- Adding small OH^- is consumed by the acid component.
- K_a measures equilibrium tendency, not reaction speed.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 20 - PRACTICE

Acid-Base Equilibria and Buffers

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
A buffer keeps pH exactly constant.	A buffer limits pH change; it does not prevent all change.	Buffer capacity is finite.
Equivalence point and endpoint are always identical.	Endpoint is indicator/instrument response; equivalence is stoichiometric.	They are related but distinct.

Guided Practice	
1. What is a buffer?	2. Give one buffer composition.
3. What does larger K_a generally indicate?	4. What happens at equivalence point?
5. What is an endpoint?	6. Does a buffer resist unlimited acid addition?
7. Name the two members of a conjugate acid-base pair.	8. What is the common-ion effect in simple terms?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. A solution that resists pH change.	2. Weak acid + its conjugate base, or weak base + conjugate acid.
3. A stronger weak acid.	4. Stoichiometric acid-base amounts have reacted.
5. Observed signal marking completion, e.g. indicator colour change.	6. No.
7. They differ by one H^+ .	8. Adding a shared ion shifts an equilibrium and can suppress ionization.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 21 - CORE CHEMISTRY

Oxidation-Reduction Reactions

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Assign oxidation numbers.
- Identify oxidation and reduction.
- Recognize oxidizing and reducing agents.

Concept	Explanation	Example
Oxidation	Loss of electrons or increase in oxidation number.	$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
Reduction	Gain of electrons or decrease in oxidation number.	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
Oxidizing agent	Species that causes oxidation and is itself reduced.	Cu^{2+} in Zn/Cu^{2+} reaction
Reducing agent	Species that causes reduction and is itself oxidized.	Zn metal

Key Rules

- Free elements have oxidation number 0.
- For a neutral compound, oxidation numbers sum to 0; for an ion, they sum to the ion charge.
- Electron loss and gain must balance in a redox reaction.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 21.

Extra Model Examples

- O is usually -2 in common compounds.
- H is usually +1 with nonmetals.
- $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ is redox.
- Oxidation and reduction always occur together.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 21 - PRACTICE

Oxidation-Reduction Reactions

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Oxidation means gaining electrons.	Oxidation means losing electrons.	Use OIL RIG: Oxidation Is Loss, Reduction Is Gain.
The oxidizing agent is oxidized.	The oxidizing agent is reduced.	It accepts electrons.

Guided Practice	
1. What is oxidation?	2. What is reduction?
3. What is the oxidation number of elemental O ₂ ?	4. What is the usual oxidation number of O in H ₂ O?
5. In $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, what is oxidized?	6. What is reduced?
7. Identify the reducing agent.	8. Why must electrons balance in redox?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Loss of electrons / increase in oxidation number.	2. Gain of electrons / decrease in oxidation number.
3. 0	4. -2
5. Zn	6. Cu ²⁺
7. Zn	8. Charge/electron conservation.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 22 - CORE CHEMISTRY

Electrochemistry

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Describe galvanic and electrolytic cells.
- Relate oxidation to anode and reduction to cathode.
- Interpret cell potential qualitatively.

Concept	Explanation	Example
Galvanic cell	Spontaneous redox reaction produces electrical energy.	battery-type cell
Electrolytic cell	External electrical energy drives nonspontaneous redox change.	electrolysis
Anode	Site of oxidation.	Zn electrode in Daniell-type cell
Cathode	Site of reduction.	Cu electrode in Daniell-type cell

Key Rules

- Oxidation occurs at the anode; reduction at the cathode in both cell types.
- Electrons flow through the external circuit from anode to cathode.
- A salt bridge or separator helps maintain charge balance in many galvanic cells.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 22.

Extra Model Examples

- An Ox, Red Cat: anode oxidation, cathode reduction.
- Galvanic cell converts chemical energy to electrical energy.
- Electrolysis converts electrical energy into chemical change.
- Positive standard cell potential corresponds to spontaneity under standard conditions.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 22 - PRACTICE

Electrochemistry

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Reduction occurs at the anode.	Reduction occurs at the cathode.	Anode is oxidation.
Electrons flow through the salt bridge.	Electrons flow through the external wire; ions move through the bridge.	Different charge carriers use different paths.

Guided Practice	
1. Where does oxidation occur?	2. Where does reduction occur?
3. In a galvanic cell, which way do electrons flow?	4. What is the role of a salt bridge?
5. What energy conversion occurs in a galvanic cell?	6. What energy conversion occurs in electrolysis?
7. What does a positive E cell indicate under standard conditions?	8. Is the anode always negative?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. anode	2. cathode
3. anode to cathode through external circuit	4. Maintains ionic charge balance and completes the circuit.
5. Chemical to electrical.	6. Electrical to chemical.
7. A spontaneous cell reaction under standard conditions.	8. No; electrode sign depends on cell type, but anode is always oxidation.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 23 - CORE CHEMISTRY

Organic Chemistry Foundations

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Explain carbon bonding and structural formulae.
- Recognize homologous series and isomerism.
- Use basic organic nomenclature.

Concept	Explanation	Example
Carbon bonding	Carbon typically forms four covalent bonds.	CH ₄
Homologous series	Family with same functional group and general pattern.	alkanes
Structural isomers	Same molecular formula, different connectivity.	C ₄ H ₁₀ isomers
Nomenclature	Names reflect carbon chain and functional group.	ethane, propene, ethanol

Key Rules

- Choose the longest relevant carbon chain for basic IUPAC naming.
- Number the chain to give key substituents or functional group low locants.
- A molecular formula does not uniquely specify structure when isomers exist.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 23.

Extra Model Examples

- meth-, eth-, prop-, but- indicate 1,2,3,4 carbons.
- -ane = alkane, -ene = alkene.
- C₂H₆ = ethane.
- C₂H₄ = ethene.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 23 - PRACTICE

Organic Chemistry Foundations

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Carbon normally forms six covalent bonds.	Carbon typically forms four covalent bonds.	It has four valence electrons in common bonding patterns.
Isomers must have different molecular formulae.	Structural isomers share a molecular formula.	They differ in connectivity.

Guided Practice	
1. How many covalent bonds does carbon usually form?	2. What prefix means two carbons?
3. What suffix identifies an alkane?	4. What suffix identifies an alkene?
5. Name C ₂ H ₆ .	6. Name C ₂ H ₄ .
7. Define structural isomerism.	8. What is a homologous series?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. 4	2. eth-
3. -ane	4. -ene
5. ethane	6. ethene
7. Same molecular formula, different connectivity.	8. A family of compounds with the same functional group and related general formula.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

LESSON 24 - CORE CHEMISTRY

Hydrocarbons

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Distinguish alkanes, alkenes, and alkynes.
- Write simple combustion and addition reactions.
- Relate saturation to bonding.

Concept	Explanation	Example
Alkanes	Saturated hydrocarbons containing only single C-C bonds.	ethane
Alkenes	Unsaturated hydrocarbons containing C=C.	ethene
Alkynes	Unsaturated hydrocarbons containing C triple C.	ethyne
Combustion	Complete combustion forms CO ₂ and H ₂ O when oxygen is sufficient.	CH ₄ + 2O ₂ → CO ₂ + 2H ₂ O

Key Rules

- Unsaturation refers to multiple carbon-carbon bonds.
- Alkenes commonly undergo addition reactions across C=C.
- Incomplete combustion can form CO and/or carbon when oxygen is limited.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 24.

Extra Model Examples

- General alkane formula: C_nH_{2n+2} for acyclic alkanes.
- General alkene formula: C_nH_{2n} for one acyclic double bond.
- Ethene can add Br₂ across the double bond.
- Complete hydrocarbon combustion produces CO₂ and H₂O.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 24 - PRACTICE

Hydrocarbons

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Alkenes are saturated.	Alkenes are unsaturated.	They contain a C=C bond.
Complete combustion of methane forms CO only.	Complete combustion forms CO ₂ and H ₂ O.	CO appears in incomplete combustion.

Guided Practice	
1. Is ethane saturated or unsaturated?	2. What functional feature defines an alkene?
3. State the general formula for acyclic alkanes.	4. Write products of complete hydrocarbon combustion.
5. What may form during incomplete combustion?	6. Name C ₂ H ₂ .
7. What type of reaction is common for alkenes?	8. Why are alkenes more reactive than analogous alkanes?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. saturated	2. C=C double bond
3. C _n H _{2n+2}	4. CO ₂ and H ₂ O
5. CO and/or carbon (soot), plus water	6. ethyne
7. addition	8. The pi component of the double bond is more reactive.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections You made, the correction, and the chemistry rule or principle.

LESSON 25 - CORE CHEMISTRY

Functional Groups and Organic Reactions

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Recognize alcohols, aldehydes, ketones, carboxylic acids, and esters.
- Use characteristic suffixes.
- Describe simple oxidation, substitution, and esterification patterns.

Concept	Explanation	Example
Alcohol	Contains -OH functional group.	ethanol
Carbonyl compounds	Aldehydes and ketones contain C=O in different positions.	ethanal; propanone
Carboxylic acid	Contains -COOH.	ethanoic acid
Ester	Contains -COO- linkage; often formed from acid + alcohol.	ethyl ethanoate

Key Rules

- Functional group largely determines typical reactions.
- Primary alcohols can be oxidized to aldehydes and further to carboxylic acids under suitable conditions.
- Esterification is reversible and commonly acid-catalysed in laboratory treatment.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 25.

Extra Model Examples

- ethanol $\rightarrow$ ethanal $\rightarrow$ ethanoic acid (oxidation sequence)
- acid + alcohol $\leftrightarrow$ ester + water
- -ol indicates alcohol
- -oic acid indicates carboxylic acid

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 25 - PRACTICE

Functional Groups and Organic Reactions

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
All oxygen-containing organic compounds are alcohols.	Different functional groups contain oxygen in different arrangements.	Structure determines class.
Esterification is always one-way.	Esterification is reversible.	Equilibrium is involved.

Guided Practice	
1. What functional group defines an alcohol?	2. What suffix is used for alcohols?
3. What group defines a carboxylic acid?	4. What suffix is used for carboxylic acids?
5. What two classes react to form an ester?	6. What is formed alongside an ester in simple esterification?
7. What can a primary alcohol oxidize to first?	8. Name one carbonyl class.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. -OH	2. -ol
3. -COOH	4. -oic acid
5. Carboxylic acid and alcohol.	6. Water.
7. An aldehyde.	8. Aldehyde or ketone.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 26 - CORE CHEMISTRY

Polymers and Biomolecules

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Distinguish addition and condensation polymerization.
- Recognize monomers and repeating units.
- Identify broad classes of biomolecules.

Concept	Explanation	Example
Polymer	Large molecule made from repeating monomer-derived units.	poly(ethene)
Addition polymerization	Unsaturated monomers add without small-molecule by-product.	ethene $\rightarrow$ poly(ethene)
Condensation polymerization	Monomers join with loss of a small molecule such as water.	polyester formation
Biomolecules	Important classes include carbohydrates, proteins, lipids, and nucleic acids.	starch, protein, DNA

Key Rules

- Polymer properties depend on monomer structure, chain length, and intermolecular interactions.
- Proteins are polymers of amino acids linked by peptide bonds.
- Polysaccharides are built from sugar monomers.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 26.

Extra Model Examples

- $n \text{ CH}_2=\text{CH}_2 \rightarrow [-\text{CH}_2-\text{CH}_2-]_n$
- Amino acids $\rightarrow$ polypeptides/proteins
- Glucose units can form polysaccharides.
- Some synthetic polymers persist in the environment.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 26 - PRACTICE

Polymers and Biomolecules

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
All polymers are synthetic.	Natural polymers include proteins, cellulose, and DNA.	Polymers occur naturally and synthetically.
Addition polymerization always produces water.	Addition polymerization typically does not release a small molecule.	Condensation polymerization often does.

Guided Practice	
1. Define monomer.	2. Define polymer.
3. Which type of polymerization commonly uses alkenes?	4. Which type often releases water?
5. What are proteins made from?	6. Name one natural polymer.
7. Name one major biomolecule class.	8. What environmental issue is associated with some plastics?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Small molecule that can join to form a polymer.	2. Large molecule with repeating units.
3. addition polymerization	4. condensation polymerization
5. amino acids	6. cellulose / protein / DNA / starch
7. carbohydrates / proteins / lipids / nucleic acids	8. Persistence and accumulation as waste/microplastics.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 27 - CORE CHEMISTRY

Analytical Chemistry: Qualitative Methods

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Use observations to identify classes of ions or substances.
- Distinguish qualitative from quantitative analysis.
- Interpret basic flame, precipitation, and gas tests.

Concept	Explanation	Example
Qualitative analysis	Determines what substances are present.	ion identification
Precipitation test	Formation of an insoluble solid can indicate particular ions.	AgCl white precipitate concept
Flame test	Some metal ions produce characteristic flame colours.	Na gives strong yellow
Gas tests	Simple observations can help identify gases.	CO ₂ turns limewater milky

Key Rules

- Use controls and clean apparatus to reduce contamination.
- One test may not be unique; confirmatory evidence is preferable.
- Record colour, precipitate, gas evolution, and other observations before interpretation.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 27.

Extra Model Examples

- Na⁺: yellow flame
- CO₂: limewater becomes milky
- H₂: burns with a small pop in a standard school test
- Precipitation depends on solubility rules.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 27 - PRACTICE

Analytical Chemistry: Qualitative Methods

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Qualitative analysis gives exact concentration.	Qualitative analysis identifies substances; quantitative analysis measures amount.	They answer different questions.
One colour observation is always definitive.	Use confirmatory tests where possible.	Interferences can occur.

Guided Practice	
1. What does qualitative analysis determine?	2. What does a precipitate indicate?
3. What flame colour is associated with sodium in common school tests?	4. What simple test is used for CO ₂ ?
5. What is a confirmatory test?	6. Why use clean apparatus?
7. Does qualitative analysis give exact molarity?	8. Name one type of qualitative observation.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. What substances or ions are present.	2. An insoluble product has formed, useful for identification.
3. yellow	4. Limewater turns milky.
5. An additional test supporting an identification.	6. To prevent contamination.
7. No.	8. Colour, precipitate, gas, flame colour, odour only where safe/approved, etc.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 28 - CORE CHEMISTRY

Analytical Chemistry: Quantitative Methods

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Relate quantitative analysis to measured amount.
- Use titration stoichiometry conceptually.
- Interpret calibration and significant figures.

Concept	Explanation	Example
Quantitative analysis	Determines how much of a substance is present.	concentration measurement
Titration	Uses a standard solution to react with an analyte.	acid-base titration
Standard solution	Solution with accurately known concentration.	0.1000 M NaOH
Calibration	Relates instrument signal to concentration or amount.	absorbance vs concentration

Key Rules

- Read burettes and volumetric glassware at eye level and record appropriate precision.
- Use the balanced reaction to convert titrant amount to analyte amount.
- Replicate measurements improve reliability and help detect outliers.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 28.

Extra Model Examples

- $n = M \times V$
- At equivalence, stoichiometric amounts have reacted.
- Calibration curve should be used within its validated range.
- Report uncertainty and significant figures appropriately.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 28 - PRACTICE

Analytical Chemistry: Quantitative Methods

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Titration volume alone gives concentration without stoichiometry.	Use volume, concentration, and reaction ratio.	Moles and coefficients are required.
More decimal places always means more accuracy.	Report only justified precision.	Instrument quality limits certainty.

Guided Practice	
1. What does quantitative analysis determine?	2. Define standard solution.
3. Write $n=M \times V$.	4. What is measured from a burette?
5. Why is a balanced equation needed in titration calculations?	6. What is a calibration curve?
7. Why repeat measurements?	8. What does equivalence point mean?

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. Amount or concentration.	2. A solution of accurately known concentration.
3. $n=M \times V$	4. Delivered volume.
5. To apply the correct stoichiometric ratio.	6. A graph relating measured signal to known concentration/amount.
7. To assess precision and reduce random error.	8. Stoichiometric reactant amounts have reacted.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 29 - CORE CHEMISTRY

Nuclear Chemistry

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Distinguish alpha, beta, and gamma radiation.
- Use half-life concept.
- Relate nuclear changes to mass-energy and stability.

Concept	Explanation	Example
Alpha decay	Emission of a helium nucleus; mass number decreases by 4, atomic number by 2.	alpha particle = 4He^{2+} nucleus
Beta minus decay	A neutron changes to a proton with electron emission; atomic number rises by 1.	beta-
Gamma emission	High-energy electromagnetic radiation; no change in A or Z.	gamma photon
Half-life	Time required for half of a radioactive sample to decay.	$N = N_0(1/2)^n$

Key Rules

- Nuclear equations conserve nucleon number and charge.
- Radioactive decay is random for individual nuclei but predictable statistically.
- Half-life is independent of initial sample size for a given nuclide under ordinary conditions.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 29.

Extra Model Examples

- After 1 half-life: 50% remains.
- After 2 half-lives: 25% remains.
- Gamma radiation is highly penetrating relative to alpha.
- Alpha is strongly ionizing but has short range.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 29 - PRACTICE

Nuclear Chemistry

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
Gamma decay changes atomic number by 1.	Gamma emission does not change A or Z.	It releases electromagnetic energy.
Half-life means all nuclei decay after that time.	Half remain on average after one half-life.	Decay is statistical.

Guided Practice	
1. What is emitted in alpha decay?	2. What happens to atomic number in beta minus decay?
3. Does gamma emission change mass number?	4. If 80 g has one half-life pass, how much remains?
5. After two half-lives, what fraction remains?	6. Is individual radioactive decay predictable exactly?
7. Which is more penetrating: alpha or gamma?	8. State one conservation rule for nuclear equations.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. A helium nucleus.	2. It increases by 1.
3. No.	4. 40 g
5. 1/4	6. No, only statistical behaviour is predictable.
7. gamma	8. Conservation of mass number/nucleon number and charge.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any corrections you made, the correction, and the chemistry rule or principle.

LESSON 30 - CORE CHEMISTRY

Environmental Chemistry and Course Review

Concepts, equations, model examples, and scientific reasoning

Learning Objectives

- Explain major chemical issues in air, water, and climate.
- Relate chemistry to pollution prevention and sustainable practice.
- Integrate core ideas from the full course.

Concept	Explanation	Example
Air pollution	Pollutants include particulates, NO _x , SO ₂ , ozone, and volatile organics.	smog chemistry
Water chemistry	pH, dissolved ions, nutrients, metals, and oxygen affect water quality.	eutrophication
Greenhouse gases	CO ₂ , CH ₄ , N ₂ O and others absorb infrared radiation and influence climate.	greenhouse effect
Green chemistry	Designs products and processes to reduce hazards and waste.	safer solvents, atom economy

Key Rules

- Distinguish pollutant concentration from toxicity; both matter for risk.
- Prevention at source is often preferable to end-of-pipe cleanup.
- Chemical decisions should consider energy use, waste, persistence, and life-cycle impacts.

Study Tip

- Read each model carefully. Then create one new example that follows the same principle as Lesson 30.

Extra Model Examples

- SO₂ and NO_x can contribute to acid deposition.
- Excess nutrients can drive algal blooms and oxygen depletion.
- Catalytic converters reduce certain vehicle emissions.
- Recycling can reduce raw-material demand but also requires energy and processing.

Worked Analysis Routine

- Step 1: Identify the chemical quantities, species, units, and conditions.
- Step 2: Select the relevant principle, equation, trend, or reaction model.
- Step 3: Show substitutions and units clearly; check chemical and mathematical consistency.
- Step 4: Interpret the result in words and test whether the magnitude or trend is reasonable.

Your Example Create chemistry example, equation, or calculation and underline the target principle.

LESSON 30 - PRACTICE

Environmental Chemistry and Course Review

Common mistakes, guided practice, mastery check, and exam strategy

Common mistake	Better form	Why?
All ozone is harmful.	Stratospheric ozone protects from UV; ground-level ozone is a pollutant.	Location and context matter.
Natural substances are always harmless.	Hazard depends on substance, dose, and exposure.	Natural origin does not guarantee safety.

Guided Practice	
1. Name one major greenhouse gas.	2. Name one pollutant linked to acid deposition.
3. What is eutrophication?	4. What is green chemistry?
5. Why is source reduction useful?	6. Is all ozone harmful?
7. Name one water-quality variable.	8. State one big idea connecting this course.

Mastery Challenge

- Write two original examples demonstrating the main idea of this lesson.
- For at least one example, show a calculation, balanced equation, or particle-level explanation.
- Check units, significant figures, chemical symbols, charge balance, and reasoning before using the quick answers.

Quick Answers	
1. CO ₂ / CH ₄ / N ₂ O	2. SO ₂ or NO _x
3. Nutrient enrichment causing excessive biological growth and often oxygen depletion.	4. Design of chemicals/processes to reduce hazards and waste.
5. It prevents pollution before it is created.	6. No.
7. pH, dissolved oxygen, nitrate, metal concentration, etc.	8. Examples: structure explains properties; amount and energy govern reactions; equilibrium and rates describe change.

Independent Practice

- A. Write three original examples using the chemistry from this lesson.
- B. Modify one model problem by changing a number, species, or condition, then solve it.
- C. Write one deliberately incorrect statement, then correct it and explain the principle.
- D. Apply the target concept to a short real-life or laboratory context in 40-60 words.

Exam Strategy

- Read the whole question before calculating; underline the required quantity and units.
- Write the governing equation or chemical principle before substituting numbers.
- After solving, check sign, units, order of magnitude, significant figures, and chemical plausibility.

Notes & Corrections

Record any mistakes you made, the correction, and the chemistry rule or principle.

Review Test 1 - Foundations & Atomic Structure

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Matter and Measurement	Classify oxygen gas: element, compound, or mixture.
2	Matter and Measurement	Classify salt water as homogeneous or heterogeneous.
3	Matter and Measurement	Convert 2500 mg to g.
4	Atomic Structure	How many protons are in oxygen (Z=8)?
5	Atomic Structure	How many neutrons are in ^{16}O ?
6	Atomic Structure	How many electrons are in Mg^{2+} (Z=12)?
7	Isotopes and Atomic Mass	What differs between isotopes of one element?
8	Isotopes and Atomic Mass	Calculate neutrons in ^{37}Cl , Z=17.
9	Isotopes and Atomic Mass	If isotope A is 20 u at 25% and B is 22 u at 75%, find the average.
10	Electron Configuration	Write the configuration of He.
11	Electron Configuration	Write the configuration of C.
12	Electron Configuration	How many electrons fit in an s subshell?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-130 word response paragraph here. Use at least four target chemistry ideas.

Review Test 2 - Periodicity, Bonding & Formulae

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Periodic Table and Trends	Which direction generally increases atomic radius down a group?
2	Periodic Table and Trends	Which direction generally increases electronegativity across a period?
3	Periodic Table and Trends	Name Group 18.
4	Chemical Bonding	Identify bond type in NaCl.
5	Chemical Bonding	Identify bond type in H2.
6	Chemical Bonding	Identify bonding in copper metal.
7	Molecular Geometry and Polarity	State the shape of CO2.
8	Molecular Geometry and Polarity	State the shape of CH4.
9	Molecular Geometry and Polarity	Is H2O polar?
10	Chemical Nomenclature and Formulae	Name Na2O.
11	Chemical Nomenclature and Formulae	Name CO2.
12	Chemical Nomenclature and Formulae	Write the formula for magnesium chloride.

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-130 word response paragraph here. Use at least four target chemistry ideas.

Review Test 3 - Mole Calculations & Stoichiometry

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Mole Concept and Molar Mass	What is Avogadro constant?
2	Mole Concept and Molar Mass	Find molar mass of CO ₂ using C=12.0, O=16.0.
3	Mole Concept and Molar Mass	Find moles in 44.0 g CO ₂ .
4	Stoichiometry	Balance: H ₂ + O ₂ -> H ₂ O.
5	Stoichiometry	What is the H ₂ :O ₂ mole ratio in the balanced equation?
6	Stoichiometry	How many mol H ₂ O form from 3 mol H ₂ if O ₂ is excess?
7	Limiting Reagents and Percent Yield	Define limiting reagent.
8	Limiting Reagents and Percent Yield	Define excess reagent.
9	Limiting Reagents and Percent Yield	What determines theoretical yield?
10	Mole Concept and Molar Mass	Find molecules in 2.0 mol H ₂ O.
11	Stoichiometry	Why must equations be balanced?
12	Limiting Reagents and Percent Yield	Can the larger-mass reactant be limiting?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-130 word response paragraph here. Use at least four target chemistry ideas.

Review Test 4 - States, Gases & Solutions

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	States of Matter and Intermolecular Forces	Which state has a fixed shape?
2	States of Matter and Intermolecular Forces	Which state fills its container?
3	States of Matter and Intermolecular Forces	Name the strongest common intermolecular force in liquid water.
4	Gas Laws	Convert 27 C to K.
5	Gas Laws	State Boyle law equation.
6	Gas Laws	State Charles law relationship.
7	Solutions and Concentration	Define solute.
8	Solutions and Concentration	Define solvent.
9	Solutions and Concentration	Calculate M for 1.0 mol in 2.0 L.
10	Solubility and Colligative Properties	Define saturated solution.
11	Solubility and Colligative Properties	Define unsaturated solution.
12	Solubility and Colligative Properties	What effect does higher pressure usually have on gas solubility?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-130 word response paragraph here. Use at least four target chemistry ideas.

Review Test 5 - Energy, Rates & Equilibrium

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Thermochemistry	What sign does Delta H have for an exothermic process?
2	Thermochemistry	What sign for endothermic?
3	Thermochemistry	Write the calorimetry equation.
4	Reaction Rates	Define reaction rate.
5	Reaction Rates	Name two factors that increase reaction rate.
6	Reaction Rates	What does a catalyst do to activation energy?
7	Chemical Equilibrium	Define dynamic equilibrium.
8	Chemical Equilibrium	Are forward and reverse rates equal at equilibrium?
9	Chemical Equilibrium	Are reactant and product concentrations necessarily equal?
10	Thermochemistry	What does c represent in $q = mc\Delta T$?
11	Reaction Rates	Why does powder react faster than a lump?
12	Chemical Equilibrium	What may happen if a product is removed?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-100 word response paragraph here. Use at least four target chemistry ideas.

Review Test 6 - Acids, Redox & Electrochemistry

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Acids, Bases and pH	Define a Bronsted acid.
2	Acids, Bases and pH	Define a Bronsted base.
3	Acids, Bases and pH	Calculate pH of $1.0 \times 10^{-3} \text{ M H}^+$ solution.
4	Acid-Base Equilibria and Buffers	What is a buffer?
5	Acid-Base Equilibria and Buffers	Give one buffer composition.
6	Acid-Base Equilibria and Buffers	What does larger K_a generally indicate?
7	Oxidation-Reduction Reactions	What is oxidation?
8	Oxidation-Reduction Reactions	What is reduction?
9	Oxidation-Reduction Reactions	What is the oxidation number of elemental O_2 ?
10	Electrochemistry	Where does oxidation occur?
11	Electrochemistry	Where does reduction occur?
12	Electrochemistry	In a galvanic cell, which way do electrons flow?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-130 word response paragraph here. Use at least four target chemistry ideas.

Review Test 7 - Organic Chemistry & Polymers

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Organic Chemistry Foundations	How many covalent bonds does carbon usually form?
2	Organic Chemistry Foundations	What prefix means two carbons?
3	Organic Chemistry Foundations	What suffix identifies an alkane?
4	Hydrocarbons	Is ethane saturated or unsaturated?
5	Hydrocarbons	What functional feature defines an alkene?
6	Hydrocarbons	State the general formula for acyclic alkanes.
7	Functional Groups and Organic Reactions	What functional group defines an alcohol?
8	Functional Groups and Organic Reactions	What suffix is used for alcohols?
9	Functional Groups and Organic Reactions	What group defines a carboxylic acid?
10	Polymers and Biomolecules	Define monomer.
11	Polymers and Biomolecules	Define polymer.
12	Polymers and Biomolecules	Which type of polymerization commonly uses alkenes?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response Write your 80-130 word response paragraph here. Use at least four target chemistry ideas.

Review Test 8 - Analysis, Nuclear & Environment

Mixed practice - complete without looking back at the lessons.

#	Topic	Question
1	Analytical Chemistry: Qualitative Methods	What does qualitative analysis determine?
2	Analytical Chemistry: Qualitative Methods	What does a precipitate indicate?
3	Analytical Chemistry: Qualitative Methods	What flame colour is associated with sodium in common school tests?
4	Analytical Chemistry: Quantitative Methods	What does quantitative analysis determine?
5	Analytical Chemistry: Quantitative Methods	Define standard solution.
6	Analytical Chemistry: Quantitative Methods	Write $n = M \times V$.
7	Nuclear Chemistry	What is emitted in alpha decay?
8	Nuclear Chemistry	What happens to atomic number in beta minus decay?
9	Nuclear Chemistry	Does gamma emission change mass number?
10	Environmental Chemistry and Course Review	Name one major greenhouse gas.
11	Environmental Chemistry and Course Review	Name one pollutant linked to acid deposition.
12	Environmental Chemistry and Course Review	What is eutrophication?

• Write a short scientific paragraph (80-100 words) using at least four chemistry ideas reviewed on this page. Include one equation, trend, or quantitative relationship where appropriate.

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Writing Task Response
Write your 80-100 word paragraph here. Use at least four target chemistry ideas.

Answer Key - Part 1

Use this section after completing the review tests independently.

Review Test 2 - Periodicity, Bonding & Formulae	
1. Down.	2. Left to right.
3. Noble gases.	4. ionic
5. covalent	6. metallic
7. linear	8. tetrahedral
9. yes	10. sodium oxide
11. carbon dioxide	12. MgCl_2

- For every wrong answer, write the corrected statement or calculation and the principle that explains it.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Error Log Write the item number, your answer, the correction, and the chemistry principle.

Answer Key - Part 2

Use this section after completing the review tests independently.

Review Test 3 - Mole Calculations & Stoichiometry	
1. $6.022 \times 10^{23} \text{ mol}^{-1}$	2. 44.0 g/mol
3. 1.00 mol	4. $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
5. 2:1	6. 3 mol
7. The reactant used up first.	8. Reactant left after the limiting reagent is consumed.
9. The limiting reagent.	10. 1.2044×10^{24} molecules
11. To satisfy conservation of atoms.	12. Yes.

- For every wrong answer, write the corrected statement or calculation and the principle that explains it.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Error Log Write the item number, your answer, the correction, and the chemistry principle.

Answer Key - Part 3

Use this section after completing the review tests independently.

Review Test 4 - States, Gases & Solutions	
1. solid	2. gas
3. hydrogen bonding	4. 300.15 K (about 300 K)
5. $P_1V_1=P_2V_2$	6. $V/T = \text{constant}$
7. Substance dissolved.	8. Medium doing the dissolving.
9. 0.50 M	10. Contains the equilibrium maximum dissolved solute.
11. Can dissolve more solute.	12. It increases it.

- For every wrong answer, write the corrected statement or calculation and the principle that explains it.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Error Log Write the item number, your answer, the correction, and the chemistry principle.

Answer Key - Part 4

Use this section after completing the review tests independently.

Review Test 6 - Acids, Redox & Electrochemistry	
1. H ⁺ donor.	2. H ⁺ acceptor.
3. 3	4. A solution that resists pH change.
5. Weak acid + its conjugate base, or weak base + conjugate acid.	6. A stronger weak acid.
7. Loss of electrons / increase in oxidation number.	8. Gain of electrons / decrease in oxidation number.
9. 0	10. anode
11. cathode	12. anode to cathode through external circuit

- For every wrong answer, write the corrected statement or calculation and the principle that explains it.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Enter Log Write the item number, your answer, the correction, and the chemistry principle.

Answer Key - Part 5

Use this section after completing the review tests independently.

Review Test 7 - Organic Chemistry & Polymers	
1. 4	2. eth-
3. -ane	4. saturated
5. C=C double bond	6. C_nH_{2n+2}
7. -OH	8. -ol
9. -COOH	10. Small molecule that can join to form a polymer.
11. Large molecule with repeating units.	12. addition polymerization

- For every wrong answer, write the corrected statement or calculation and the principle that explains it.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Ever Log Write the item number, your answer, the correction, and the chemistry principle.

Answer Key - Part 6

Use this section after completing the review tests independently.

Review Test 8 - Analysis, Nuclear & Environment	
1. What substances or ions are present.	2. An insoluble product has formed, useful for identification.
3. yellow	4. Amount or concentration.
5. A solution of accurately known concentration.	6. $n = M \times V$
7. A helium nucleus.	8. It increases by 1.
9. No.	10. CO ₂ / CH ₄ / N ₂ O
11. SO ₂ or NO _x	12. Nutrient enrichment causing excessive biological growth and often oxygen depletion.

- For every wrong answer, write the corrected statement or calculation and the principle that explains it.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Error Log Write the item number, your answer, the correction, and the chemistry principle.

REFERENCE

Periodic Table, Constants and Common Ions

A compact exam reference for the course.

Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H																He	
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe

Selected Atomic Masses (approx.)

- H 1.008; C 12.01; N 14.01; O 16.00; Na 22.99; Mg 24.31; Al 26.98; Si 28.09; P 30.97; S 32.06; Cl 35.45; K 39.10; Ca 40.08; Fe 55.85; Cu 63.55; Zn 65.38; Ag 107.87.

Useful Constants

- Avogadro constant $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.
- Ideal gas law: $PV = nRT$. Use an R value consistent with pressure and volume units.
- At 25 °C for dilute aqueous solutions: $\text{pH} + \text{pOH} = 14$.
- 1 atm approximately 101.3 kPa; 1 L = 1000 mL.

Common Ions

- Cations: H^+ , Na^+ , K^+ , Ag^+ , NH_4^+ , Mg^{2+} , Ca^{2+} , Zn^{2+} , Cu^{2+} , Fe^{2+} , Fe^{3+} , Al^{3+} .
- Anions: Cl^- , Br^- , I^- , OH^- , NO_3^- , HCO_3^- , SO_4^{2-} , CO_3^{2-} , PO_4^{3-} .

Solubility Reminders (introductory)

- All nitrates and common alkali-metal salts are generally soluble in water.
- Many chlorides are soluble, with important exceptions such as AgCl .
- Many carbonates are poorly soluble except alkali-metal and ammonium carbonates.
- Always use the solubility rules supplied by your course or exam board when precision is required.

REFERENCE

Chemistry Quick Guide

Core equations, patterns, and checks for final revision.

Topic	Key relation / reminder
Density	density = mass / volume
Moles	$n = m/M$; $N = n \times N_A$
Solutions	$M = n/V$; dilution $M_1V_1 = M_2V_2$
Gases	$PV=nRT$; use kelvin in gas-law calculations
Calorimetry	$q = mc\Delta T$
Rate	change in concentration or amount / time
Equilibrium	dynamic: forward rate = reverse rate; K changes with temperature
Acid-base	$pH = -\log[H^+]$; $[H^+] = 10^{-pH}$
Percent yield	actual / theoretical $\times 100\%$
Redox	oxidation = electron loss; reduction = electron gain
Electrochemistry	anode = oxidation; cathode = reduction
Nuclear	after n half-lives, fraction remaining = $(1/2)^n$

Final Exam Checklist

- Balance equations before stoichiometry.
- Write units on every line of quantitative work.
- Convert Celsius to kelvin when required by gas-law relationships.
- Distinguish strong/weak from concentrated/dilute.
- Check charge balance in ions and oxidation numbers.
- Use significant figures consistent with the data.
- For equilibrium, distinguish rate, position, and equilibrium constant.
- For organic chemistry, identify the functional group before predicting reactions.

Course Completion

- You have completed 30 chemistry lessons, 8 review tests, answer-key checking, and the final reference section. Revisit your error log before any major examination.

Final Study Note

Final study note: active: solve, explain, check, correct, and repeat.